CHEE 591: Investigation into the Exxon Valdez Oil Spill

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CHEE 591 Environmental Bioremediation

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Abstract

The 1989 Exxon Valdez oil spill released approximately 11 million gallons of oil onto over 2000 kilometers of shoreline in Prince William Sound, Alaska. This disaster initiated one of the largest and most complex remediation efforts in global history. The unique hydrogeological features of Alaska shaped the behavior of contaminants and largely influenced the methods used to clean up the spill. Early response relied heavily on mechanical methods to clean up the spill, while bioremediation efforts were being researched on their viability. As natural attenuation and population growth of indigenous microorganisms became evident, it was clear that bioremediation would become the primary strategy. Ecological impacts were severe and caused several political and legal changes in the oil industry in the short term. In the long term, recovery trends show a slow progression into contaminant levels pre-spill. This paper outlines the geological context of the spill and the remediation timeline and concludes with providing recommendations on which remediation should have been conducted.

1. Introduction

The Exxon Valdez Oil Spill was a catastrophic accident with a heroic response and proved that biostimulation is an effective environmental cleanup method. The spill took place on March 24th, 1989, in the waters of the Gulf of Alaska in the North of the Pacific Ocean, about 12-25 miles from the shores of Prince William Sound. The contamination was caused by an oil tanker that ran aground and collided with the Bligh reef under the water (Daily Dose). Official reports from the National Oceanic and Atmospheric Administration (NOAA) say that the crash was caused by crew tiredness, navigational errors from malfunctioning radar, and Exxon's failure to maintain proper staffing. Journalists suggest that the true cause was a drunken captain who gave control of the ship to an unlicensed crew member (Daily Dose). In any case, the crash gave way for 11 million gallons of oil to be spilled into the ocean, wreaking havoc on marine life and contaminating the beach's soil and groundwater. The cleanup took place over many decades as different physical chemical degradation and bioremediation methods were employed, including biostimulation, which was a critical tool for degrading the oil.

Numerous geological formations were impacted by the spill, including ocean water, groundwater onshore, and sediments of the ocean floor and beach. Hydrocarbons in oil are hydrophobic and

formed layers on top of the ocean due to lower density. This controlled initial spreading behaviour and slowed vertical movement of the spill. Eventually, enough of the oil diffused into the water and spread throughout. The sediments of the intertidal vadose zone were impacted, with amplified effects because of the tide cycle shifting the location of the saturated zone. In a study by Qioana Guo in 2010, it was shown that the water table of the beach was higher than the interface between the surface and lower layers, which prevented much of the oil from penetrating the lower layers of sediment. A higher water table meant the oil was trapped in the upper layer and penetrated little into the lower sediments. The lower layer had a lower hydraulic conductivity compared to the upper gravel layer which slowed the flow of contaminants into the saturated layer. According to a 2010 study, "due to the two-layer structure of the beaches in Prince Willam Sound... the upper layer's permeability is 100 to 1000 times that of the lower layer" (Li and Boufadel). The permeability contrast limited the downward migration of the oil. Additionally, the freshwater recharge into the upper soil had different effects than saltwater on the spread of contaminants. Qioana Guo showed that freshwater recharge causes contaminated plumes to move downwards first. It opposes the flow towards shore and prevented oil from migrating deeper into the sediment (Guo 2010). Importantly, some oil did reach the lower layer of soil in the saturated zone. It was not until the 2010s that an effort was made to clean up this portion of the contamination.

The site characteristics of the Prince William Sound beaches played a role in how the correct remediation method was determined. The most visible characteristic was the vast spread of the plume because of its hydrophobic properties slowing downward spread and increasing outward spread. The oil spill area grew to more than 3000 square miles, showing it was impossible to be physically contained (NOAA). The concentration of petroleum hydrocarbons ranged from roughly 0.2 – 1.5 ppb.

The next site characteristic of note is the average salinity of the water near the beach. It was determined to be 24.2 grams salt per liter of water (Guo, 2010). This had an impact on what microbes can grow. Another factor is the high ammonia to nitrate ration in contaminated zones (Boufadel). This signals that the conditions were anaerobic in the regions. The bacteria had likely used up oxygen to degrade the first available oil. Also, wind conditions in the open waters help to evaporate the volatile components. According to Wolfe, 1994, "~20% of Prudhoe Bay crude

oil are most likely to evaporate". This, coupled with high degradation rates for the volatile components, meant that a significant portion of the spill was resolved by the wind. The light hydrocarbons had half lives in air of approximately ½ day, which meant that within a week 99% of the light hydrocarbons were degraded.

Inside of the lower sediment layer, there were low levels of nutrients. Boufadel et al. used chromatography in 2010 to determine that the concentrations of pore water nutrients in the contaminated sediments were less than 0.5 mg/L nitrogen and less than 0.04 mg/L phosphorus. Under these conditions, the biodegradation rate is significantly diminished. Optimal levels of the nutrients for these sediment conditions were greater than 2 mg/L nitrogen and 0.2 mg/L phosphorus.

The temperature of the ocean impacted the activity of microorganisms, the types of microorganisms present, and the chemical properties of the contaminants. The Arctic climate meant that surface water temperatures were cold, ranging between 5 to 0 degrees Celsius. This is generally suboptimal for microbial activity, slowing biodegradation rates. Also, the oil is slow to degrade and disperse naturally at colder temperatures, making these conditions unfavourable for cleanup when compared to spills in warmer climates. (Nelson, 2019). These conditions drastically decreased the treatment window for biodegradation. (Pritchard 1992)

2. Methodology

2.1 Abiotic methods of oil cleanup

In the early stages of the spill, a controlled burn was attempted to degrade the oil slick. Two ships attached by a fire-resistant boom were used to round up oil, before ignition. Although a fire was lit, weather effects extinguished the flame and the cleanup crew decided they needed a new approach (EPA.gov).

The second method of abiotic cleanup was a mechanical removal of the oil using boats attached to skimmers and booms. Logistical issues caused by the remote location caused a delay in the arrival of equipment, which meant that the oil had already spread far by the time skimmers could try to contain the oil. In a 1989 report to the President of the United States, the (U.S.) National Response Team wrote, "Initial industry efforts to get equipment on scene were unreasonably

slow, and once deployed the equipment could not cope with the spill". Mechanical removal methods were employed continuously throughout the remediation.

Next, the response crew made attempts to add chemical dispersant to the oil spill to separate the light non-aqueous phase layer into more easily degraded pieces. Efforts ceased when the Coast Guard decided that this was not effective because of the low amounts of dispersants used and delay in application. (EPA)

As mentioned in the introduction, the wind helped in the degradation by volatilizing the light hydrocarbons where they were quickly degraded. (Wolfe). The final abiotic method employed was the application of hot, high-pressure water to remove oil from the beach and rocks on the shoreline. This approach was unsuccessful and wrecked the habitats of indigenous organisms.

2.2 Bioremediation of Beaches

Following the oil spill, several applications of bioremediation to reduce petroleum hydrocarbon concentrations were attempted.

Immediately after the spill, microorganism concentration of oil degraders increased 10-000 times over, indicating a strong potential for bioremediation. Thus, Pritchard and a team of scientists went through rigorous testing and determined that an oleophilic, slow releasing granulated fertilizer (Inipol EAP22) would best act as a nutrient source for oil-degrading microbes. These fertilizers released nutrients over 24-72 hours and then continuously for the next few weeks. They adhered well to the oiled cobble in the beach matrix, making them an excellent choice of fertilizer, and were spread along the surface of the beaches. Pritchard found that the addition of fertilizer roughly doubled biodegradation rates.

Other bioremediation attempts began in 1989 on the shoreline. A second fertilizer was introduced alongside Inipol EAP22. Customblen, a slow-release granulated fertilizer with composition (28% N, 3.5% P). Approximately 50,000 kg of nitrogen and 5,000 kg of phosphorus were applied to shorelines over the summers of 1989-92. The goal was to increase the nutrient availability for microbes. This was an early trial of biostimulation.

As part of the 1989-1992 biostimulation efforts, a study done by Dr. Bragg explained how monitoring wells were created in 1990 to get accurate data on the changes in hydrocarbon levels

and nutrient composition. Three wells were added in three beaches that were split into fertilized and non-fertilizing halves. Three sampling wells per section were dug 70cm deep for samples to be taken before adding the fertilizers, and repeatedly for 109 days after. The water samples analyzed for nutrients: NH_4^+, NO_3^-, and total organic nitrogen. Also. The dissolved oxygen content, salinity. pH and temperature were measured. The soil samples taken were analyzed for total heterotroph and hydrocarbon degrader populations as well as chemical composition and amount of oil in soil.

Even 20 years after the spill, several beaches around Prince William Sound were still contaminated with polycyclic aromatic hydrocarbons (PAH) among other petroleum hydrocarbons. Prior to intervention, the degradation rate of the PAHs was estimated to be around 1% annually. Michel Boudfadel, a professor studying the spill, implemented a study where they injected these sites with hydrogen peroxide to bio-stimulate microorganisms by adding an oxygen source. The hydrogen peroxide was injected along with nutrients and diluted with sea water. This mixture was released at controlled intervals into the lower layer of beaches (the upper layer's permeability is 100-1000x more than lower layers, and solutions applied to the surface, i.e. upper layer, tend to dilute and "wash out to sea more rapidly than they can be transported into the contaminated layer."). Hydrogen peroxide was selected due to its solubility in water and decomposition into O2 and H2O. The nutrients in the mixture were nitrate (obtained from lithium and sodium nitrate), and phosphate (obtained from sodium tripolyphosphate). To analyze the effectiveness of this injection, TPAH, dissolved oxygen, and nutrient concentrations were measured over time (mg X / g total extractable material). The study showed that bioremediation was successful, and accelerated biodegradation rates by 12-60-fold. The study was not conducted without an environmental cost however, as there were significant disruptions of the beaches and a need to access forested areas inwards of the beach. Furthermore, the harsh weather risks equipment damage and malfunction, which could result in spillage of other contaminants / substances into the environment.

3. Results and Discussion

3.1 Abiotic Methods

The initial controlled burn, although conceptually effective for surface oil, was not successful. Burning must occur shortly after the oil spill, thus the delay and harsh weather conditions rendered the controlled burn ineffective.

Mechanical removal of the oil also produced limited results. As mentioned before, the equipment deployment time was ‘unreasonably slow’, indicating several logistical failures before the skimmers and booms were even on the water. Furthermore, the rough Alaskan waters proved difficult to navigate and thus mechanical recovery was deemed too slow but were still carried out continuously.

The addition of chemical dispersant was attempted as well. 4,000 gallons of dispersant were available and were applied via helicopter and dispersant bucket. Unfortunately, the water was not strong enough to properly mix the dispersant, leading to their inefficacy and thus the United States Coast Guard discontinuing their use.

Hot, high-pressure water was used to try and physically remove the oil from the shoreline. This method is often viable, but in this instance, the high-pressure water streams indirectly harmed the indigenous wildlife. Additionally, scientist P.H. Pritchard hypothesizes that physical washing could have forced the oil to a greater depth thus making bioremediation a slightly more difficult path to pursue. Regardless, bioremediation was still viable as detailed in section 3.2, especially due to the high porosity of the beaches.

Natural attenuation seemed to be the most effective way to remove oil, as much of the lighter hydrocarbons rose to the surface of the water and were volatilized by the wind, and indigenous microbe populations skyrocketed immediately after the spill.

3.2 Bioremediation Methods

Pritchard’s initial application of granulated, oleophilic and slow releasing fertilizer saw biodegradation rates increase by around 2 fold, with oil residue loss rates more than doubling. Pritchard also measured hopane, a stable hydrocarbon which is difficult to biodegrade, and the hexane-insoluble fraction, a collection of biodegradation byproducts, before and after the

fertilizer injection. He found that the hopane concentrations remained relatively stable, but the hexane-insoluble fraction grew six times, indicating that indeed, the oil *was* being metabolized.

Other early biostimulation trials added different fertilizers, agreeing with Pritchard's conclusions about the viability of biodegradation, especially in nutrient-limited areas.

Several years later, deeper contamination was addressed by Boufadel, who enriched the deeper layers of sediment by the injection of hydrogen peroxide alongside nutrients and seawater. This method dramatically increased biodegradation rates relative to natural attenuation and confirmed that deeper sediments environments were still harsher and had minimal oxygen.

However, these approaches did not come without challenges. Boufadel's injection method required access to forested areas inwards of the beach and complex piping networks. Furthermore, the harsh weather conditions of Alaska posed additional risk to equipment, and the injection of water could potentially spread contaminants further.

Regardless, the benefits of bioremediation seem to outweigh the negatives.

3.3 Results of Remediation Efforts

Overall, the combined remediation strategies employed at the Exxon Valdez Oil Spill site demonstrated mixed, but ultimately positive results. Since the spill was so large, and many studies were conducted, it is difficult to quantify the exact success of the spill.

Plenty of non-mechanical methods worked well. Natural attenuation and indigenous microbial activity were highly effective for lighter, easier to degrade hydrocarbons. Biostimulation reliably accelerated the breakdown of other petroleum hydrocarbons, and fertilizer formulations were chosen because of their ability to exist within the soil matrix. Deep layer oxygen stimulation also proved to be successful in enabling microorganisms to degrade the hydrocarbons that were found deeper into the soil matrix.

Many of the mechanical methods were ineffective and failed due to a variety of reasons. Controlled burning was unsuccessful due to weather constraints and the time restrictions. Physical removal and dispersants faced many logistical delays and lack of material, and high pressure washing harmed ecosystems and potentially expanded the oil matrix.

3.4 Ecological and Human Effects

The spill had severe short term ecological consequences. 250,000 seabirds, 2,800 sea otters, 900 Bald Eagles and 300 Harbor Seals were killed following the spill (Oceana, 2020) . Other marine life got affected as well. Even 4 years later, pink salmon embryos were still getting harmed by the oil.

The spill devastated local commercial fishing communities, and the long-term legal battles that followed reshaped the oil industry and their responsibilities in oil spill prevention. Cleanup operations themselves were hazardous, and four human deaths were linked to the cleanup efforts. The disaster inflicted roughly $300 million in losses on the 32,000 residents living in the area who depend on fishing.

3.5 Long Term Outcomes

In the long term, oil residues persisted in the deeper layers of the beach and shoreline matrices, primarily with polycyclic aromatic hydrocarbons. Regardless, decades of remediation efforts and natural attenuation have resulted in many of the beaches recovering to similar conditions prior to the spill.

The spill itself triggered major regulatory changes, including the Oil Pollution Act of 1990, stronger Coast Guard emergency response planning, and standardized protocols for remediation.

Conclusions and recommendations

The Exxon Valdez spill was unparalleled in size for its time. Remediation was a colossal task requiring a rapid response and a variety of methods. The response team did an impressive job. Nevertheless, several recommendations are presented that are likely to have been beneficial to clean up.

Hot, high-pressure washing of the shoreline should have been avoided because, as NOAA reports, this method disturbed local organisms and disrupted natural bioremediation. It scoured away the silty soils that served as habitat for microbes, algae, plants, and animals, effectively sending these fine sediments into the sea and destroying the conditions necessary for microbial activity. NOAA also notes that “high-pressure water streams may drive oil into the beach sediments where oil may become trapped or further contaminate clean areas,” which is exactly

what occurred on Prince William Sound beaches, where deeper sediment layers later showed very slow biodegradation rates. Had the hot-water pressure washing been reserved for cleaning boats instead of shorelines, the hydrocarbons on the beaches would likely have been degraded more efficiently through natural microbial processes.

The results presented by Dr. Boufadel showed that there were many inefficiencies in the biostimulation of the beaches. In multiple trials, nitrate utilization rates differed between sites. Sites with low nitrate concentration used up the nitrogen quickly after injection and could have benefitted from a larger injection. It would have been more efficient to study the microbes present in the soil so that the optimal amounts of nutrients could have been injected. Techniques such as polymerase chain reaction could have been used more to get a better understanding of the microbial communities in the soil. Also, known hydrocarbon degraders could have been added to the soil in an act of biostimulation. These microbes should have been selected so that they would use the nutrients injected, reducing waste, and accelerating biodegradation.

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